

Solution Requirements

Maximum Marks: 2 Marks

Date

15 March 2026

Team ID

Batch 2023-27 | ACOE

Project Name

Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

Functional Requirements

Functional requirements define the specific behaviours and features the Rising Waters system must implement.

Req. ID	Requirement	Priority
FR-01	System shall accept 6 meteorological inputs via a web form	High
FR-02	System shall validate all inputs and return meaningful error messages	High
FR-03	System shall scale inputs using the pre-fitted StandardScaler (scaler.save)	High
FR-04	System shall load XGBoost model (floods.save) and return binary classification	High
FR-05	System shall display a dedicated Flood Chance result page (red theme)	High
FR-06	System shall display a dedicated No Flood Chance result page (green theme)	High
FR-07	System shall log every prediction with timestamp to predictions_log.csv	Medium
FR-08	System shall display full prediction history on the History page	Medium
FR-09	System shall show model accuracy comparison chart on the Home Dashboard	Medium

Non-Functional Requirements

Req. ID	Category	Requirement
NFR-01	Performance	Prediction response time shall be under 2 seconds

Req. ID	Category	Requirement
NFR-02	Accuracy	XGBoost model test accuracy shall be $\geq 95\%$
NFR-03	Availability	System shall be available 24/7 via IBM Cloud hosting
NFR-04	Usability	Responsive design, accessible on mobile and desktop browsers
NFR-05	Security	Session-based result passing; no raw model file exposure
NFR-06	Scalability	Dockerized deployment, ready for horizontal scaling
NFR-07	Maintainability	Modular code: separate utils/, model/, templates/ directories